Infosys — Aptitude Test 4

Questions 76–100 of 100 · 25 questions

Question 76. There are N number of railway stations. Each station issues tickets for every other station. Some stations are added. Now they have to issue 46 more tickets. Give the No. of stations after and before added.

I n f o s y s

Solution:
The number of tickets issued from N number of stations = nP2
Let x be the number of stations added.
New number of tickets = n+xP2
n+xP2 – nP2 = 46
(n+x)(n+x-1) – n(n-1) = 46
The equation obtained cannot be solved, Hence, substitute values given in the options in the above equation to get the answer.
In this case the answer is:
n = 11; x = 2

Question 77. 7 members have to be selected from 12 men and 3 women , such that no two women can come together . In how many ways we can select them ?

Solution:
There should not be two women or more in the selection. That means only one woman should be selected from the three women.
Ways of selecting 1 woman = 3C1
The remaining six persons should be selected from 12 men in 12C6 ways.
Total ways of selection
= 12C6 x 3C1
= 2772 ways.

Question 78. In how man was team of four can be formed from four boys and three girls such that at least one boy and one girl should be there?

Solution:
Under the given conditions, the team can be formed in either of the following ways.
3 boys and 1 girl, or
2 boys and 2 girls, or

1 boy and 3 girls.

Ways of selecting 3 boys and 1 girl
= 4C3 x 3C1 = 4 x 3 = 12
Ways off selecting 2 boys and 2 girls
= 4C2 x 3C2 = 6 x 3 = 18
Ways of selecting 1 boy and 3 girls
= 4C1 x 3C3 = 4 x 1 = 4
Total ways of selection
= 12 + 18 + 4 = 34

2, 3, 5, 7, 11.
See the below table to identify, how many results give these numbers as the sum.

I n f o s y s

Question 79. If a die has 1,6 and 3,4 and 2,5 opposite each other how many such dies can be made.

Results of dice 2

Solution:
Take any one side as reference, lets say 1.
Side opposite to 1 is always 6.
Imagine that you are facing the side 1.
To the right of side 1, it may be 3 or 4
And to the top of side 1 it may be 2 or 5.
In each case we have 2 possibilities.
So the number of ways the dice can be printed is
= 2 + 2 = 4

The number of shaded boxes = number of expecting results
Probability = 15/36

Question 80. 15 tennis players take part in a tournament. Every player plays twice with each of his opponents. How many games are to be played?

Solution:
Ways of selecting one spade = 13C1
Ways of selecting one heart = 13C1
Ways of selecting 2 cards = 52C2
Probability of selecting one heart and one spade

Solution:
If each player has to play with the other 14 players, the number of games would be
15C2 = (15 x 14)/(2 x 1) = 105
Since each player has to play twice with every other player, the number matches

will get doubled.
Total number of matches = 105 x 2 =
210

=
=

= 13/102

Question 81. Two dice are thrown simultaneously. What is the probability that the sum of the numbers shown on the two dices will be a prime number?

Solution:
The number of ways of arranging 5 boys and 5 girls in a circle, where 5 boys are sitting together is:
= 5!(10 – 5)! = 5! x 5!
Number of ways of arranging 10 persons around a circle is:
= (10 – 1)! = 9!

Solution:
The minimum sum obtained by throwing two dice 2 and the maximum sum is 12.
The prime numbers in this range are:

Results of dice 1
1 2 3 4 5 6

6 5 4 3 2 1

13C1 x 13C1

13 x 13

52C2

1326

Probability of the above case is:
= (5! x 5!)/9!
= (5 x 4 x 3 x 2 x 1)/(9 x 8 x 7 x 6)
= 5/126

Question 82. Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart, is:

Question 83. 5 boys and 5girls sit around a circular table. What is the probability that 5 boys are sitting together?

Question 84. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

Solution:
The numbers divisible by 3 in this range are
= 3, 6, 9, 12, 15, 18
The numbers divisible by 5 in this range are
= 5, 10, 15 and 20
15 is occurring in both groups. So we neglect one 15 and take the other.
Leaving one 15, the expected number of results are 9
Probability = 9/20.
Note:
A confusion might arise when you think that 15 is divisible by both 3 and 5, so it will not come in the result.
Compare the question with logic gate.
or = OR gate; “either or” = EXOR gate

Question 85. There is a school were 60% are girls and 35% of the girls are poor. Students are selected at random, what is the probability of selecting a poor girl out of total strength?

Solution:
Assume there are 100 students in the school.
Number of girls = (60/100)100 = 60
Number of poor girls
= (35/100)60 = 21
Probability of selecting one poor girl
= 21/100

Question 86. A bag contains 64 balls of 8 different colors. There are eight of each color (including red). What is the least number of balls one must pick, without looking ,to be sure of selecting 3 red balls?

Solution:
The minimum number of balls to be selected to get one red ball is 57 because, all the balls that have been selected before may be of colors other than red.
58th draw will assure you of second red ball.
59th draw will assure you of third red ball.

Therefore the number of draws = 59

I n f o s y s

Question 87. The main line train starts at 5.00AM and the harbor line train starts at 5.02AM.Each train has the frequency of 10 minutes. If a guy goes in the morning at a random time what is the probability of he getting main line train?

Solution:
The duration between two main line trains is 10 minutes.
In this 10 minutes, if a passenger reaches in the first two minutes, he will get a harbor line train.
In the remaining 8 minutes, he will get a main line train.
The probability is:
= 8/10 = 0.8

Question 88. What is the probability of A/B to be an integer when A=2x3y and B=2l3m and all of x, y, l, m are positive integers?

Solution:
A and B are four digit numbers.
100 different values can be substituted for x and y pair. They are
[(0,0), (0,1),(0,2),….(5,6),…(9,9)]
Similarly 100 pairs are possible for (l,m)
[(2x3y),(2l3m)] is a pair and only if x = l and y = m, A/B will be an integer.
For example, if (x,y) = (4,5), then A = 2435
Therefore, (l,m) = (4,5) ■B = 2435
Total pairs of (A,B) = 100 x 100 = 10000
Pairs with (x = l),(y = m) = 100
Probability = 100/10000
=1/100

252 = 625; S = 6; H = 2; E =5;
(S+H+E)= 13

Question 89. You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and

it comes up heads. The probability that the other face is tails is?

I n f o s y s

Question 90. One card is drawn from a pack of 52 cards. What is the probability, that it is a spade or ace?

Solution:
Probability
of
selecting
one
spade
=13/52
Probability of selecting one ace = 4/52
Probability of selecting a spade or ace
= 13/52 + 4/52 – 1/52
= 16/52
(1/52) should be subtracted because one of the ace will be from spade, which is already considered in 13/52, when choosing a spade.

Question 91. If (HE)^H=SHE, where the alphabets takes the values from (0-9) & all the alphabets are single digit then find the value of (S+H+E).

+ U S
S R

Solution:
In the expression (HE)^H = SHE, HE is a two digit number and SHE is a three digit number.
Let us start with assuming numbers for H.
If H = 0, then (HE)^H = 1, which does not give the required expression.
If H = 1, a two digit number raised to power 1 will give a two digit number.
We will not get the required expression.
If H ≥3, a two digit number raised to 3 or more will give a number which has more than 3 digits.
Only possible value is: H = 2

P E
A C E

Since A = 2; (9+S) = 12, where we get A= 2 and 1 carry over for 9 in USSR.
SO, S = 12 – 9 = 3.

+ 9 3 S
8

E = 0; P = 1; U = 9; S = 3; C = 7; A = 2;
R = 8

P+E+A+C+E = 1 + 0 + 2 + 7 + 0 = 10

1

9
S
2

U S A

=

+ 9 S
S
8

1 0 A C
0

1
1

1
1

9
3
2

9 S
2

+ 9 3
3
8

=

1 0 2
7
0

1 0 2 C
0

Question 92. USA+USSR=PEACE;P+E+A+C+E=?

Solution:
This is a question from crypt arithmetic. Here each letter is assigned with one of the digit from 0 to 9.
Corresponding digit for some of the letters can be identified easily through basic concepts.
For ex, In the above question, a three digit number is added with a four digit number to get a five digit number. So, the first number from left of the five digit number will definitely be 1.
P = 1.
Only when a three digit number is added to a four digit number which starts with

9, we will get a 5 digit number.
U = 9.
When a three digit number is added with a four digit number and we get a five digit number, the second digit from left of the five digit number will be 0.
E = 0.
A + R = 0
Let us start with assuming values for A and R
If A = 2; R = 8, there will be one carry over for S + S

Solution:
Since one of the face is head, it should be either HH coin or HT coin.
The probability of getting tail in the other face is (1/2).

Question 93. EVER+SINCE=DARWIN, then D+A+R+W+I+N=?

Solution:
D = 1; A = 0; S = 9, Similar to previous question.
Another method of solving this kind of questions is by using a tabulation.
We have to write the numbers which are sure and then assume values for others.
Here we start with assuming value for E

A
D
S
E
I
R
N
C
V
W

If E = 2, I>7 to get carry

0
1
9
2
8
0
X

0
1
9
2
9
X

If E = 3,I>6 to get carry

0
1
9
3
7
0
X

0
1
9
3
8
1
X

0
1
9
3
9
X

If E = 4,I>5 to get carry

0
1
9
4
6
0
X

0
1
9
4
7
1
X

0
1
9
4
8
2
6
4
X

0
1
S
4
9
X

If E = 5,I>4 to get carry

0
1

9
5
5
X

0
1
9
5
6
1
X

0
1
9
5
7
2
7
X

0
1
9
5
8
3
8
X

If there is a carry from

V+N, then

0
1
9
5
7
3
8
2
6
4
√

D = 1; A= 0; R = 3; W = 4; I = 7; N = 8
D+A+R+W+I+N = 1+0+3+4+7+8 =
23

Question 94. CROSS+ROADS=DANGER. FIND D+A+N+G+E+R.

Solution:
For sure, D = 1.
R is an even number.
And E = S + D

D
R

S
E
O
A
G
N
C

I n f o s y s

If R= 4; S = 2

If O = 5

1
4
2
3
5
6
1
X

1
4
2
3
5
7
2
X

1
4
2
3
5
8
3
X

1
4
2
3
5
9
4
X

If O = 6

1
4
2
3
6
5
1
X

1
4
2
3
6
6
X

1
4
2
3
6
7
3
X

1
4
2
3
6
8
4
X

1
4
2
3
6
9
5
1
X

If O = 7
1
4
2
3
7
5
2
X
1
4
2
3
7
6
3
X
1
4
2
3
7
7
X

1
4
2
3
7
8
5
2
X
1
4
2
3
7
9
6
2
X
If O = 8
1
4
2
3
8
5
3
X
1
4
2
3
8
6
4
X
1
4
2
3
8
7
5
3
X
1
4
2
3
8
8
X
1
4
2
3
8
9

7
3
X
If O = 9
1
4
2
3
9
5
4
X
1
4
2
3
9
6
5
4
X
1
4
2
3
9
7
6
4
X
1
4
2
3
9
8
7
4
X
1
4
2
3
9
9
X
If R = 6; S = 3

If O = 2
1
6
3
4
2
5
7
8

9
√

D =1; A =5; N =8; G =7; E =4; R =6
D+A+N+G+E+R = 31

Question 95. EAT+EAT+EAT=BEET if t=0 then what will the value of TEE+TEE.

Value of I cannot be one. So, we can substitute 6 in the place of I. I = 6.
1+I+I = 1+6+6 = 13. There is a carry over to 7+F.
1+7+F = 16
F = 8.

I n f o s y s

Solution:
T = 0.

T
A
E
B
E
E
T

0
1
3
0
9
3
0
X

0
2
6
1
8
6
0
X

0
3
9
2
7
9
0
X

S = 0; I = 6; N = 7; E = 5
S+I+N+E = 0+6+7+5 = 18

0
4

2
0
7
2
0
X

0
5
5
X

Question 96. NINE+FINE = WIVES. Find S+I+N+E. E = 5 and V = 3.

Solution:
S= E+E= 5+5= 0(taking only the unit digit).
W = 1.

1

N
I N
E

X + Y + Z = 14

N
I N
5

+
F
I N
E

+
F
I N
5

Question 97. XYZ+XYZ+XYZ=ZZZ. Find X+Y+Z

0
6
8
2
5
8
0
X

Solution:
Z+Z+Z= Z in the unit place. Z = 0 or 5.
Z cannot be 0. So, Z = 5; Z+Z+Z = 15.
1 is carry over to Y+Y+Y.

(1+Y+Y+Y) ends in 5. 1+8+8+8 = 25.
Y = 8.
2 is the carry over to Z+Z+Z.
2+Z+Z+Z = 5
2+3Z = 5
Z = 1.

0
7
1
0
5
1
0
X

0
8
4
1
4
4
0
√

T = 0; E = 4
TEE+TEE = 044+044
88

Question 98. COCA+COLA= OASIS. Find O+A+S+I+S .

■

1
I V E
S

1
I 3 5
0

Solution:
O = 1.
A+A = S. Two times a same number will always give an even number.
So, S is an even number.
So, O+O = S ■1+1 = 2.
S = 2; O = 1.
A+A = 2.
A should be either 1 or 6 to give S = 2.
A cannot be equal to 1, because O = 1.
So, A = 6
A+A = 12. there is a carry over to C+L.

There should be a carry over for I+I to get 3.
If N = 2, we will get 2+2+1 = 5, but

there is no carry over.
If N = 7, we get 7+7+1 = 15 with a
carry over. So, N = 7.

1 1

7
I 7
5

+
F
I 7
5

1
I 3 5
0

1 1 1

7 6 7
5

+
8 6 7
5

1
6 3 5
0

2 1

1 8 5

X Y Z

1 8 5

X Y Z

■

+ 1 8 5

+ X Y Z

5 5 5

Z Z Z

Since O=1 and A = 6, first two digits of
OASIS is 16.
C+C should be equal to 16.
C = 8.
1+C+L = I.
There is no carry over from the above
expression to O+O.
So 1+C+L should be less than 10.
1+C+L = 1+8+L < 10
9 + L < 10
So, L = 0
I = 1+8+0 = 9
O+A+S+I+S = 1+6+2+9+2 = 20

Question 99. MOON+SUN=PLUTO. Find P+L+U+T+O

Solution:
P = 1.
A four digit number is added to get a
three
digit
number.
The
four
digit
number will be number starting with 9.
M = 9; and L = 0.
O is an even number because it is twice
of N.

L
P
M
O
N
U
T
S

If O = 2

0
1
9
2
1
X

0
1
9
2
6
3
7
11
X

If O = 4 and V=3,5,6,8

0
1
9
4
2
3
7
9
X

0
1
9
4
7

3
8
9
X

0
1
9
4
7
5
0
X

0
1
9
4
7
6
1
X

0
1
9
4
7
8
3
13
X

if O = 6 and V = 2,3,4,5,7

I n f o s y s

0
1
9
6
3
2
8
6
X

0
1
9
6
8
2
9
X

0
1
9
6
8

3
0
9
X

0
1
9
6
8
4
1
X

0
1
9
6
8
5
2
8
X

0
1
9
6
8
7
4
10
X

If O = 8 and V = 2,3,…

0
1
9
8
4
2
0
X

0
1
9
8
4
3
1
X

0
1
9
8
4
5

3
6
√

P = 1; L = 0; U = 5; T = 3; O = 8
P+L+U+T+O = 1+0+5+3+8 = 17

Question 100. BANANA+GUAVA = ORANGE. Find O+R+A+N+G+E

Solution:
O = B+1; E = A+A; N = 1+A+A
E is always even number.
There should be a carry to A+A= N and
A+G=R

I n f o s y s

O = 3; R = 1; A = 4; N = 9; G = 6; E = 8.
O+R+A+N+G+E = 3+1+4+9+6+8

= 31

Infosys was
launched in 1981 by

Mr. Narayana
Murthy with six
other colleagues on
a mere $250 (10000
INR) borrowed from

his wife.